

AAKASH SANGANI

Data Scientist | ML and LLM Systems

Mumbai, India | aakash.sangani2410@gmail.com | [linkedin.com/in/aakash-sangani-3149a11a7](https://www.linkedin.com/in/aakash-sangani-3149a11a7) | github.com/aakash2410

SUMMARY

Data scientist with 3+ years across pharma analytics, fintech experimentation, and AI governance. Builds and evaluates ML and LLM systems, from Bayesian inference and fairness audits to agentic analytics products. Author of two open-source AI evaluation tools.

TECHNICAL SKILLS

Machine learning: Bayesian inverse reinforcement learning, LSTM, XGBoost, SVM, gradient boosting, ensembles, SMOTETomek, SHAP, IBM AIF360, scikit-learn, TensorFlow

LLM and agentic AI: RAG, agentic AI, MCP servers, LLM evaluation and red-teaming

Data and experimentation: Python, SQL, pandas, PostgreSQL, Apache Spark, A/B testing

Infrastructure: AWS, Docker, CI/CD gates with SARIF reporting

AI governance frameworks: EU AI Act, NIST AI RMF, ISO 42001, GDPR/DPDP, HIPAA, OWASP ASI, MITRE ATLAS

PROFESSIONAL EXPERIENCE

Kalpa Impact

Feb 2026 – present

AI Consultant

Mumbai

- Digital public infrastructure and AI governance consulting for UNDP, DPGA, Gates Foundation, and Co-Develop.
- Sole PM and builder on an offline RAG system, in collaboration with the Government of India's Bhashini.
- Collaborated with the Principal Scientific Adviser's Office to write a section in "AGI Governance" on AI incident reporting mechanisms India can adopt, based on the EU AI Act and California's SB-53.

ZS Associates

Sep 2023 – Jan 2026

Business Technology Solutions Associate, promoted to Senior Associate

Pune

- Launched an agentic AI analytics product for natural-language querying of sales-performance metrics.
- Owned incentive-compensation product logic for a Fortune 100 pharma client, delivered at 100% accuracy.
- Led a team of 4. Recognised as a top 1% performer firm-wide; promoted to Senior Associate within 8 to 10 months of joining.

Bright Money

Jul 2022 – Feb 2023

Product Analyst Intern

Bangalore

- Launched a referral programme that saved \$120K in customer-acquisition cost over 5 months.
- Built A/B testing infrastructure and SQL dashboards.

SELECTED PROJECTS

Inverse-Objective Regulator (IOR)

github.com/aakash2410/Inverse-Objective-Regulator

Designed and built a black-box AI-safety auditing system; sole author of the accompanying arXiv paper

- Infers a deployed AI agent's revealed objective from its tool-call traces via Bayesian inverse reinforcement learning, scores divergence from its declared purpose, and emits regulator-legible evidence mapped to the EU AI Act, OWASP, MITRE ATLAS, and NIST AI RMF.
- Released an open 8-agent benchmark spanning agentic safety failure modes (reward hacking, power-seeking, sycophancy, privacy leakage, safety-skipping); a deterministic open cross-encoder critic recovers planted objective divergence on 6/8 agents (mean $r = 0.55$), beating a frontier LLM-as-judge (mean $r = 0.08 \rightarrow 0.43$) with near-zero variance and no API cost.
- Sole author of the accompanying arXiv system paper and full Python implementation (Bayesian IRL, LLM/decision-model judges, reproducible gym, JSON/SARIF reporting).

Credit Equity Audit of India (NSSO 77th Round)

github.com/aakash2410/CreditAudit-India

Fairness audit of household credit access and allocation

- Two-stage, Heckman-style deep learning model: stage 1 predicts credit market participation, stage 2 predicts formal versus informal credit allocation. Female-headed households had about 59% of the probability of market entry of male-headed households (disparate impact 0.59).
- Applied IBM AIF360 adversarial debiasing and SHAP to detect proxy discrimination, reaching near-ideal disparate impact at ~73.2% accuracy. Documented limits of the design (self-reported survey data, collider bias risk, SHAP as a model diagnostic, not causal evidence).

rag-compliance-auditor

github.com/aakash2410/MCP_RAG_Compliance

Sole-authored open-source package, published on PyPI

- MCP server that certifies RAG and AI pipelines against the EU AI Act, GDPR/DPDP, HIPAA, SEBI/RBI, and NIST AI RMF across 7 audit dimensions. Issues RS256-signed JWT certificates, exposes 8 MCP tools, and runs as a CI/CD gate with SARIF reporting.

Baby Cry Reason Detection

2022 – 2023

Final-year project (team of 3)

- Classified the cause of an infant's cry from a 457-sample audio dataset using temporal, spectral, cepstral, and prosodic features (MFCCs, spectral contrast, pitch). Handled class imbalance with SMOTETomek; final model was a soft-voting SVM + KNN ensemble, deployed in a Flutter app with a feedback loop for retraining.

Personality Prediction with CV and Social Media Analysis

2022

Team project (team of 4)

- A 7-layer LSTM reached 0.82 – 0.85 accuracy on the Big Five traits (myPersonality dataset, 250 users, 9,917 status updates), against 0.45 – 0.61 for SVM, Logistic Regression, Gradient Boosting, and XGBoost.

PUBLICATION

Sangani, A., Doshi, P., & Sawant, V. "Unusual Social Media Behavior Detection Using Distributed Data Stream Mining." In *Practical Data Mining Techniques and Applications*. Auerbach Publications (CRC Press), 2022 – 23. Clickstream-based bot and hijack detection using per-cluster Very Fast Decision Trees on Apache Spark, with concept-drift handling.

EDUCATION

D. J. Sanghvi College of Engineering, University of Mumbai

2019 – 2023

B.Tech, Information Technology. CGPA 8.90 / 10. Merit scholarship covering 100% of tuition

Coursework: Probability & Statistics (O, 93%), Statistical Analysis (A+, 81%), Artificial Intelligence (A+, 84%), Advanced Data Structures (O, 95%), Design and Analysis of Algorithms (A+, 83%), Discrete Structures (A+, 84%).